

Logic Pro Template Build Guide — MPD-CLA Session v3.0

Step-by-step instructions for building the pre-routed Logic Pro session template. This creates a ready-to-mix session with 8 busses, 4 FX sends, and 20 audio tracks, all loaded with MPD-CLA presets.

Prerequisite: All 25 presets must be built and saved first. See BUILD-GUIDE.md.

Quick Reference

TRACKS (20)	BUSSES (8)	FX SENDS (4)	
Track 1-4 : Drums	---> Bus 1: DRUMS	--> Bus 7: MIX	
Track 5-6 : Bass	---> Bus 2: BASS	--> Bus 7: MIX	
Track 7-10: Guitars	---> Bus 3: GUITARS	--> Bus 7: MIX	
Track 11-14: Keys	---> Bus 4: KEYS	--> Bus 7: MIX	
Track 15-18: Vocals	---> Bus 5: VOCALS	--> Bus 7: MIX	
Track 19-20: FX/SFX	---> Bus 6: FX	--> Bus 7: MIX	
	Bus 7: MIX	--> Bus 8	Sen
	Bus 8: MASTER	--> Output 1-2	Send
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Step 1: Create a New Empty Project

1. File > New (or Cmd+N)
 2. Choose **Empty Project**
 3. When prompted to create a track, click **Cancel** — we'll create everything manually
 4. Set project tempo (120 BPM default, change as needed)
 5. Set sample rate: **48 kHz** (recommended)
 6. Set bit depth: **24-bit**
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Step 2: Create the 8 Busses

Create busses in reverse order (Master first) so routing targets exist before they're needed.

Bus 8: MASTER

1. Mix > Create New Auxiliary Channel Strip
2. Set input to **Bus 8**
3. Set output to **Output 1-2** (Stereo Out)
4. Name: MASTER
5. Color: **Red**
6. Load preset: Open Library (Y) > User Patches > MPD-CLA > MPD-MASTER-BUS
7. Set fader to **0 dB**

Bus 7: MIX

1. Mix > Create New Auxiliary Channel Strip
2. Set input to **Bus 7**
3. Set output to **Bus 8**
4. Name: **MIX**
5. Color: **Orange**
6. Load preset: MPD-BUS-MIX
7. Set fader to **0 dB**

Busses 1-6: Instrument Groups

Create each bus in order:

	Bus Input	Output	Name	Color	Preset
1	Bus 1	Bus 7	DRUMS	Blue	MPD-BUS-DRUMS
2	Bus 2	Bus 7	BASS	Purple	MPD-BUS-BASS
3	Bus 3	Bus 7	GUITARS	Green	MPD-BUS-GUITARS
4	Bus 4	Bus 7	KEYS	Yellow	MPD-BUS-KEYS
5	Bus 5	Bus 7	VOCALS	Pink	MPD-BUS-VOCALS
6	Bus 6	Bus 7	FX	Gray	MPD-BUS-FX

For each:

1. Mix > Create New Auxiliary Channel Strip
 2. Set input to the corresponding Bus number
 3. Set output to **Bus 7 (MIX)**
 4. Name and color per the table above
 5. Load the preset from Library > User Patches > MPD-CLA
 6. Set fader to **0 dB**
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Step 3: Create the 4 FX Send Returns

These are Aux channel strips that receive signal from send knobs on tracks and busses.

	Send Input	Output	Name	Color	Preset
1	Bus 9	Bus 7	SHORT VERB	Light Blue	MPD-FX-SHORT-VERB
2	Bus 10	Bus 7	LONG VERB	Cyan	MPD-FX-LONG-VERB
3	Bus 11	Bus 7	SLAP DLY	Teal	MPD-FX-SLAP-DLY
4	Bus 12	Bus 7	LONG DLY	Mint	MPD-FX-LONG-DLY

For each:

1. Mix > Create New Auxiliary Channel Strip
2. Set input to the send bus (Bus 9/10/11/12)
3. **Set output to Bus 7 (MIX)** — FX returns sum into the mix bus, not the master
4. Name and color per the table above
5. Load the preset from Library > User Patches > MPD-CLA
6. Set fader to **0 dB**

Important: FX send presets are 100% wet. The dry/wet balance is controlled by the send level on each source track, not on the FX return.

Step 4: Create Audio Tracks

Create 20 audio tracks with default routings. Tracks can be added or removed later — this is the starting template.

Drums (Tracks 1-4)

Track	Name	Output	Input
1	Kick	Bus 1	Input 1
2	Snare	Bus 1	Input 2
3	Toms	Bus 1	Input 3
4	OH/Room	Bus 1	Input 4

Bass (Tracks 5-6)

Track	Name	Output	Input
5	Bass DI	Bus 2	Input 5
6	Bass Amp	Bus 2	Input 6

Guitars (Tracks 7-10)

Track	Name	Output	Input
7	GTR L	Bus 3	Input 7
8	GTR R	Bus 3	Input 8
9	Acoustic	Bus 3	Input 9
10	GTR Clean	Bus 3	Input 10

Keys (Tracks 11-14)

Track	Name	Output	Input
11	Piano	Bus 4	—
12	Synth	Bus 4	—
13	Pad	Bus 4	—
14	EP	Bus 4	—

Vocals (Tracks 15-18)

Track	Name	Output	Input
15	Lead Vox	Bus 5	Input 1
16	BG Vox L	Bus 5	Input 2
17	BG Vox R	Bus 5	Input 3

FX/SFX (Tracks 19-20)

Track	Name	Output	Input
19	Riser/FX	Bus 6	—
20	Impact/SFX	Bus 6	—

For each track:

1. Track > New Audio Track (or Option + Cmd + N)
2. Set output to the appropriate bus (not Stereo Out)
3. Rename the track
4. Color-code to match its bus group

Loading Mixing Presets on Tracks

Individual tracks get frequency-domain mixing presets based on their content:

Track	Recommended Preset	Why
Kick	MPD-CLA-SUB	Fundamental below 80 Hz
Snare	MPD-CLA-HI-MID	Crack and body 500 Hz-2 kHz
Toms	MPD-CLA-LO-MID	Body and attack 80-500 Hz
OH/Room	MPD-CLA-AIR	Shimmer and air 8-20 kHz
Bass DI	MPD-CLA-SUB	Sub-bass fundamental
Bass Amp	MPD-CLA-LO-MID	Harmonics and midrange grit
GTR L/R	MPD-CLA-HI-MID	Presence and bite
Acoustic	MPD-CLA-HIGH	Pick attack and clarity 2-8 kHz
GTR Clean	MPD-CLA-HI-MID	Clean presence
Piano	MPD-CLA-HI-MID	Body and clarity
Synth	MPD-CLA-HIGH	High harmonic content
Pad	MPD-CLA-LO-MID	Warmth and body
EP	MPD-CLA-HI-MID	Bell-like presence
Lead Vox	MPD-CLA-HI-MID	Vocal clarity and presence
BG Vox	MPD-CLA-HIGH	Brightness, sit behind lead
Harmonies	MPD-CLA-HIGH	Complementary brightness
Riser/FX	MPD-CLA-AIR	Open, shimmery character
Impact/SFX	MPD-CLA-SUB	Low-end weight

Note: These are starting points. Swap presets based on the actual content of each track.

Step 5: Configure Sends on All Tracks

Add 4 sends to every audio track and busses 1-6. All sends start at **-inf (off)** by default.

Adding Sends

For each audio track and busses 1-6:

1. In the Mixer (Cmd + 2) or Channel Strip inspector, click an empty Send slot
2. Route to the FX send bus:
 - Send 1 > **Bus 9** (SHORT VERB)
 - Send 2 > **Bus 10** (LONG VERB)
 - Send 3 > **Bus 11** (SLAP DLY)
 - Send 4 > **Bus 12** (LONG DLY)
3. Set send mode to **Post-Fader** (default, recommended)
4. Leave all send levels at **-inf (off)**

Recommended Starting Send Levels

These are suggested starting points — adjust to taste:

Source Bus	SHORT VERB	LONG VERB	SLAP DLY	LONG DLY
DRUMS	-12 dB	Off	-18 dB	Off
BASS	Off	Off	Off	Off
GUITARS	-10 dB	-14 dB	Off	-16 dB
KEYS	-12 dB	-10 dB	Off	Off
VOCALS	-8 dB	-12 dB	-14 dB	-12 dB
FX	Off	Off	Off	Off

Tip: Apply sends at the bus level rather than per-track for consistent group reverb. Only use per-track sends when a specific track needs different FX treatment than its group.

Step 6: Arrange the Mixer

Organize the mixer for fast navigation:

Recommended Mixer Order (left to right)

[Tracks 1-4] [Bus 1] | [Tracks 5-6] [Bus 2] | [Tracks 7-10] [Bus 3] |
[Tracks 11-14] [Bus 4] | [Tracks 15-18] [Bus 5] | [Tracks 19-20] [Bus 6]
[FX 1-4] | [Bus 7: MIX] | [Bus 8: MASTER]

Track Stacks (Optional but Recommended)

Group tracks into Summing Stacks for cleaner arrangement:

1. Select tracks 1-4, then Track > Create Track Stack > **Summing Stack**
2. Name the stack "DRUMS" — this creates a folder in the arrangement

3. Repeat for each group: BASS (5-6), GUITARS (7-10), KEYS (11-14), VOCALS (15-18), FX (19-20)

Note: If using Summing Stacks, the stack's output bus becomes the group bus automatically. Verify routing after creating stacks.

Step 7: Set Channel Strip I/O Defaults

Headroom Settings

Channel	Fader	Pan
All audio tracks	-6 dB	C (center)
Busses 1-6	0 dB	C
FX returns 1-4	0 dB	C
Bus 7 (MIX)	0 dB	C
Bus 8 (MASTER)	0 dB	C

Starting audio tracks at **-6 dB** provides headroom before the bus stage.

Pan Defaults (Adjust Per Song)

Track	Pan
Kick	C
Snare	C
Toms	-20 to +20
OH/Room	-40/ +40 (stereo)
Bass DI	C
Bass Amp	C
GTR L	-50
GTR R	+50
Acoustic	+25
Lead Vox	C
BG Vox L	-30
BG Vox R	+30

Step 8: BCF2000 Setup (Optional)

If using a Behringer BCF2000 hardware controller:

1. Connect BCF2000 via USB
2. Logic Pro > Settings > Control Surfaces > Setup
3. Add the BCF2000 (should auto-detect)
4. See [spec/bcf2000.md](#) for detailed encoder mapping

Quick BCF2000 Workflow

- Select a track/bus in Logic
 - The BCF2000's 8 rotary encoders map to Smart Controls knobs 1-8
 - Knob assignments follow the selected channel's preset automatically
 - Switch between tracks to control different presets with the same 8 knobs
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Step 9: Save as Template

1. Verify all routing (see Step 10 first)
2. File > Save as Template...
3. Template name: MPD-CLA Template
4. Collection: **My Templates** (or create a new collection)
5. Click **Save**

The template is saved to:

~/Music/Audio Music Apps/Project Templates/MPD-CLA Template.logicx

Using the Template

When starting a new project:

1. File > New from Template
 2. Select **MPD-CLA Template**
 3. All busses, FX sends, and routing are pre-configured
 4. Import/record audio to the appropriate tracks
 5. Swap mixing presets on tracks as needed for the specific song
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Step 10: Verification Checklist

Run through these checks before saving the template.

Routing Verification

- [] All audio tracks output to their assigned bus (1-6), NOT to Stereo Out
- [] Busses 1-6 all output to Bus 7 (MIX)
- [] Bus 7 (MIX) outputs to Bus 8 (MASTER)
- [] Bus 8 (MASTER) outputs to Output 1-2 (Stereo Out)
- [] FX returns (Bus 9-12) all output to Bus 7 (MIX)
- [] No track or bus outputs directly to Stereo Out (except Bus 8)

Signal Flow Test

1. Play test audio through one track per bus group
2. Verify signal appears on Bus 1-6 meters
3. Verify signal sums into Bus 7 (MIX)
4. Verify signal passes through Bus 8 (MASTER)

5. Verify signal reaches Output 1-2

FX Send Test

1. Raise Send 1 (SHORT VERB) on a vocal track to -6 dB
2. Verify signal appears on the SHORT VERB return (Bus 9)
3. Verify the reverb return sums into Bus 7 (MIX)
4. Repeat for each FX send
5. Reset all test sends to -inf

Preset Verification

- [] Each bus has its correct preset loaded (check Smart Control labels)
- [] Each FX return has its correct preset loaded
- [] Bus 7 labels: INPUT / LOW / HIGH / GLUE / WARM / WIDTH / AIR / OUTPUT
- [] Bus 8 labels: INPUT / LOW / HIGH / GLUE / LOUD / WIDTH / CEILING / OUTPUT
- [] FX returns show effect-specific knob labels (DECAY, TIME, etc.)
- [] Sweep each knob on Bus 7 and Bus 8 — verify parameter responds

Gain Staging Check

1. Play pink noise or test audio at -18 dBFS through all tracks simultaneously
 2. Bus 1-6 meters should read approximately -12 to -6 dBFS
 3. Bus 7 (MIX) should read approximately -6 to 0 dBFS
 4. Bus 8 (MASTER) should not clip with all presets at default settings
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Troubleshooting

"No signal on bus"

- Check the track's output assignment in the Mixer — it may have reverted to Stereo Out
- Verify the Aux channel strip's input matches the expected bus number

"FX send has no effect"

- Confirm the FX return Aux input matches the send bus number (Bus 9/10/11/12)
- Check that the FX preset is 100% wet (not a dry/wet mix issue)
- Verify send level is raised above -inf on the source track

"Preset won't load on bus"

- Bus/Aux channels use the same preset library as audio tracks
- Ensure you're loading from **Patches** (Library, Y key) not Channel Strip Settings
- Smart Controls only persist in .patch format

"Clicking/distortion at bus stage"

- Reduce individual track faders — summing many tracks at 0 dB causes clipping
- Use the -6 dB track fader default as your starting point

- The DRIVE knob on bus presets adds input gain — start at 0
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File Locations Reference

```
~/Music/Audio Music Apps/  
├─ Project Templates/  
│   └─ MPD-CLA Template.logicx          <-- This template  
├─ Channel Strip Settings/MPD-CLA/  
│   ├── MPD-CLA-SUB.cst                 <-- Mixing presets  
│   ├── MPD-CLA-LO-MID.cst  
│   ├── MPD-CLA-HI-MID.cst  
│   ├── MPD-CLA-HIGH.cst  
│   ├── MPD-CLA-AIR.cst  
│   ├── MPD-BUS-DRUMS.cst               <-- Bus presets  
│   ├── MPD-BUS-BASS.cst  
│   ├── MPD-BUS-GUITARS.cst  
│   ├── MPD-BUS-KEYS.cst  
│   ├── MPD-BUS-VOCALS.cst  
│   ├── MPD-BUS-FX.cst  
│   ├── MPD-BUS-MIX.cst  
│   ├── MPD-FX-SHORT-VERB.cst           <-- FX send presets  
│   ├── MPD-FX-LONG-VERB.cst  
│   ├── MPD-FX-SLAP-DLY.cst  
│   ├── MPD-FX-LONG-DLY.cst  
│   ├── MPD-MASTER-BUS.cst             <-- Mastering presets  
│   ├── MPD-MASTER-SUB.cst  
│   ├── MPD-MASTER-LO-MID.cst  
│   ├── MPD-MASTER-HI-MID.cst  
│   ├── MPD-MASTER-HIGH.cst  
│   ├── MPD-MASTER-AIR.cst  
│   ├── MPD-MASTER-STREAM.cst  
│   └─ MPD-MASTER-MULTI.cst  
└─ Patches/MPD-CLA/  
    └─ (same filenames with .patch extension)
```